

PARKTOWN BOYS' HIGH SCHOOL



Department of Mathematics

June Examination Paper 2

3 June 2024

Subject	Mathematics
Grade	12
Marks	150
Time	3 hours

Examiner

Ms H Gouws

Moderator

Mrs S Rowell

INSTRUCTIONS

Read the following instructions carefully before answering the questions.

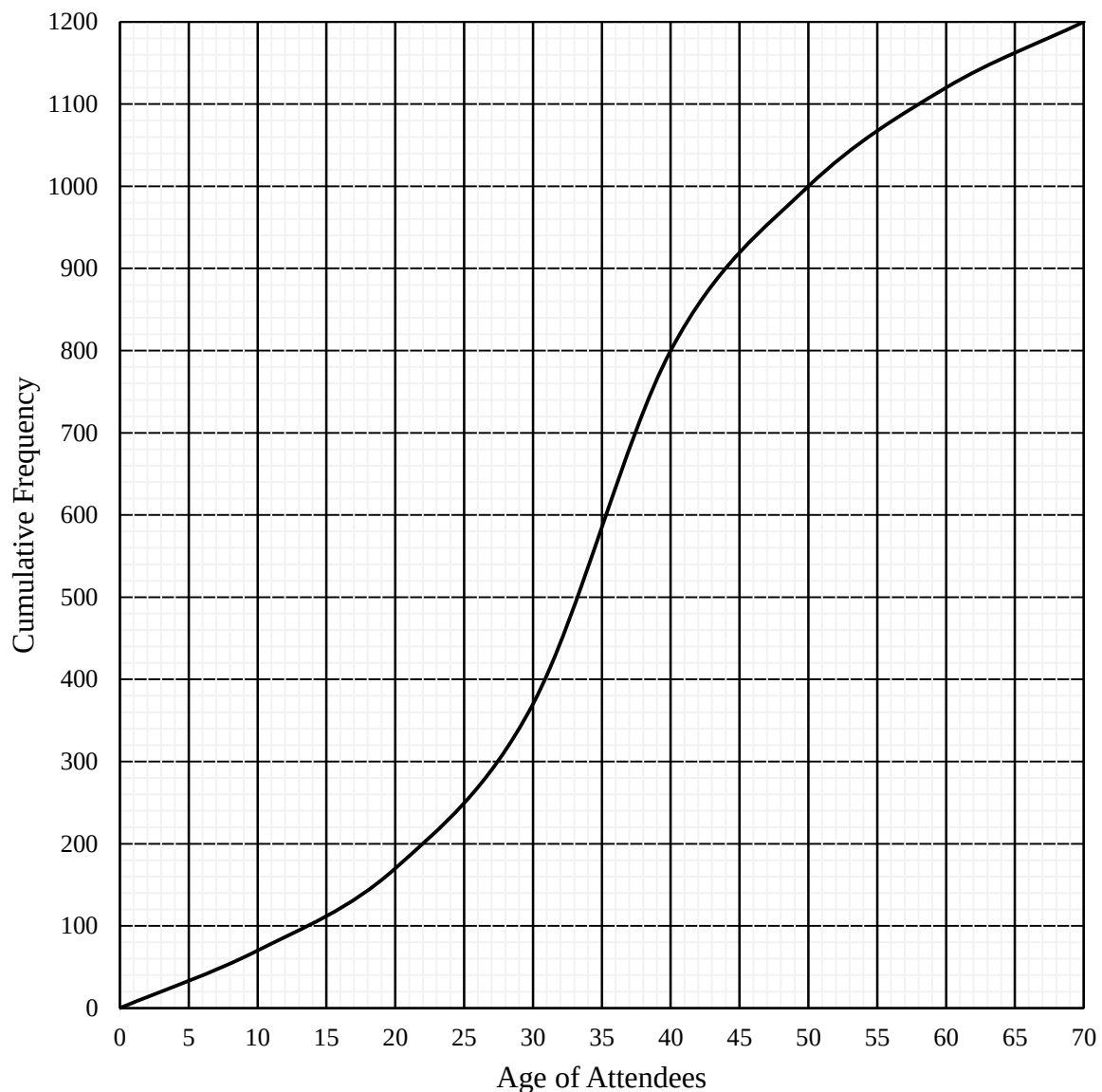
1. This question paper consists of 10 questions, 14 pages and a formula sheet. You may detach the formula sheet for your use.
Please ensure that your question paper is complete.
2. Answer ALL the questions in the answer books provided. Additional space is provided at the end of the answer book.
3. Clearly show ALL calculations, diagrams, graphs, etc. that you have used in determining your answers.
4. Answers only will NOT necessarily be awarded full marks.
5. You may use an approved scientific calculator (non-programmable and non-graphical), unless stated otherwise.
6. If necessary, round off answers to TWO decimal places, unless stated otherwise.
7. Diagrams are NOT necessarily drawn to scale.
8. Use the mark allocation to guide the length of your solution.
9. Write neatly and legibly.

QUESTION 1

The number of people, per age group, who attended the Parktown Centenary Family Fun Day is recorded in the table below:

Age Group	Number Attended	Cumulative Frequency
$0 < x \leq 10$	70	70
$10 < x \leq 20$	100	170
$20 < x \leq 30$	200	370
$30 < x \leq 40$	A	C
$40 < x \leq 50$	B	D
$50 < x \leq 60$	120	1120
$60 < x \leq 70$	80	1200

This data is represented in the cumulative frequency graph below:



- 1.1 Write down the values of A, B, C and D in the table above. (4)
- 1.2 Use the cumulative frequency graph above to estimate the median age of the people who attended the Parktown Centenary Family Fun Day. (2)
- 1.3 If the minimum age was 3 and the maximum age was 68, draw a box and whisker plot to represent the ages of attendees. (5)
- 1.4 Comment on the distribution of the ages of people who attended the event. Give a reason for your answer. (2)
- 1.5 The organisers decide to charge children under 12 half price for the 2024 event. What effect would this have on the distribution of the data? (2)

[15]

QUESTION 2

At the Grey Festival the shoulder strength and number of push-ups done per day by a random sample of twelve eighteen-year-old first team rugby players were tested. The results are recorded in the table below:

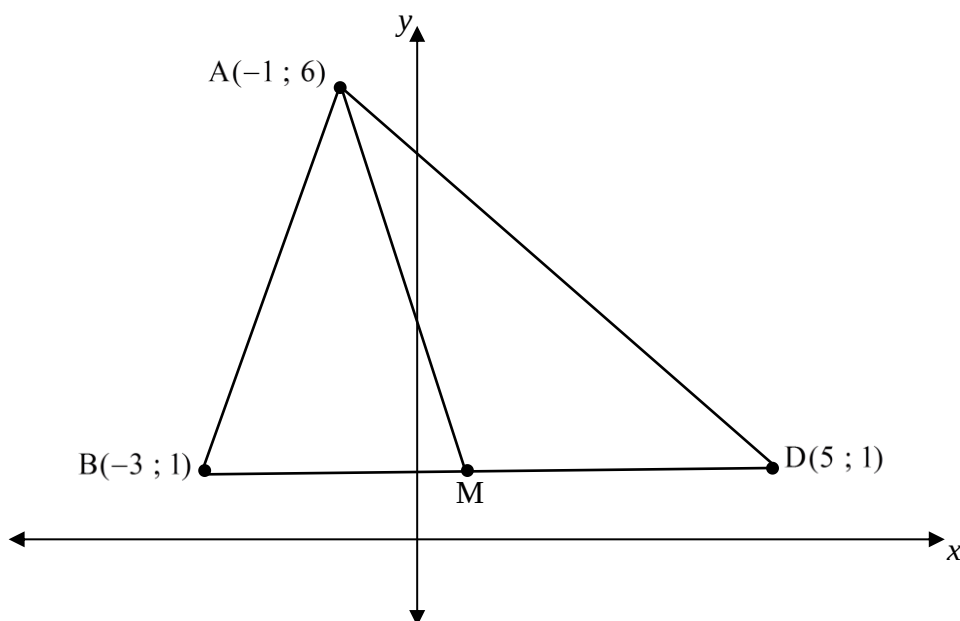
Number of daily push-ups (x)	27	165	56	156	99	40	70	68	20	105	56	158
Shoulder strength (y)	20	98	36	95	72	28	60	60	15	78	40	95

- 2.1 Calculate the correlation coefficient. (2)
- 2.2 Describe the relationship between the number of push-ups and the shoulder strength of an eighteen-year-old first team rugby player. (2)
- 2.3 Determine the equation of the least squares regression line. (2)
- 2.4 Use the regression line to predict the shoulder strength of an eighteen-year-old first team rugby player who does a 120 push-ups per day. (2)
- 2.5 Mr Clay does 280 push-ups per day. Should the regression line calculated in QUESTION 2.3 predict his shoulder strength? Motivate your answer. (2)

[10]

QUESTION 3

In the diagram below, $\triangle ABD$ with $A(-1 ; 6)$, $B(-3 ; 1)$ and $D(5 ; 1)$ are drawn. AM is a median of the triangle.

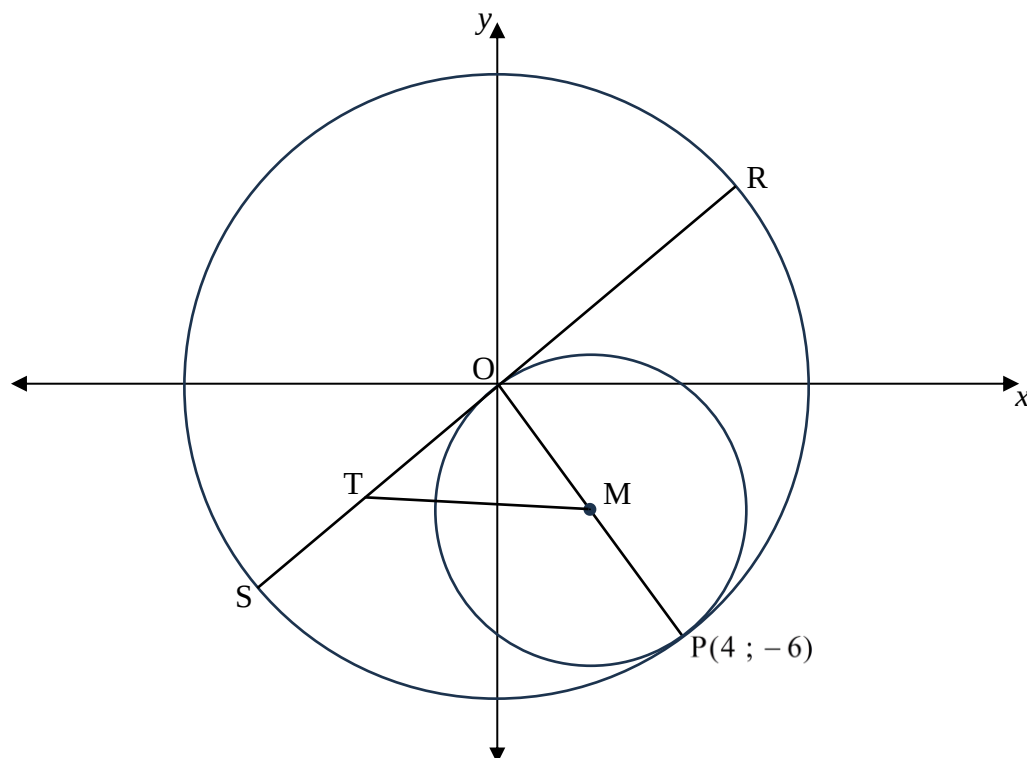


- 3.1 Calculate the gradient of AD . (2)
- 3.2 If $ADEB$ is a parallelogram, determine the coordinates of E . (4)
- 3.3 Determine the equation of median AM . (4)
- 3.4 If A , B and $C(-5 ; t)$ are collinear, determine the value of t . (4)
- 3.5 Determine the size of $\hat{BAD}$. (4)

[18]

QUESTION 4

In the diagram below, a circle with centre at the origin passes through $P(4 ; -6)$. PO is a diameter of the smaller circle with centre M . RS , the diameter of the larger circle, is a tangent to the smaller circle at O . T is a point on OS and MT has been drawn.



- 4.1 Determine the equation of
- 4.1.1 the larger circle. (2)
 - 4.1.2 the smaller circle. (4)
 - 4.1.3 RS . (3)
- 4.2 Determine the coordinates of R . (5)
- 4.3 If $MT = 5$ units, determine the length of OT . (3)
- [17]**

QUESTION 5

5.1 Simplify each of the following without the use of a calculator:

5.1.1 $\cos(75^\circ + \theta)\cos(15^\circ + \theta) + \sin(75^\circ + \theta)\sin(15^\circ + \theta)$ (3)

5.1.2 $\tan(-\theta)\sin(90^\circ + \theta) + \frac{\sin(360^\circ + 2\theta)}{2\cos(360^\circ - \theta)}$ (6)

5.2 Consider the identity $\frac{\sin 4x \cos 2x - 2 \cos 4x \sin x \cos x}{\tan 2x} = \cos^2 x - \sin^2 x$.

5.2.1 Prove this identity. (5)

5.2.2 For which values of $x \in [0^\circ ; 180^\circ)$ is this identity undefined? (3)

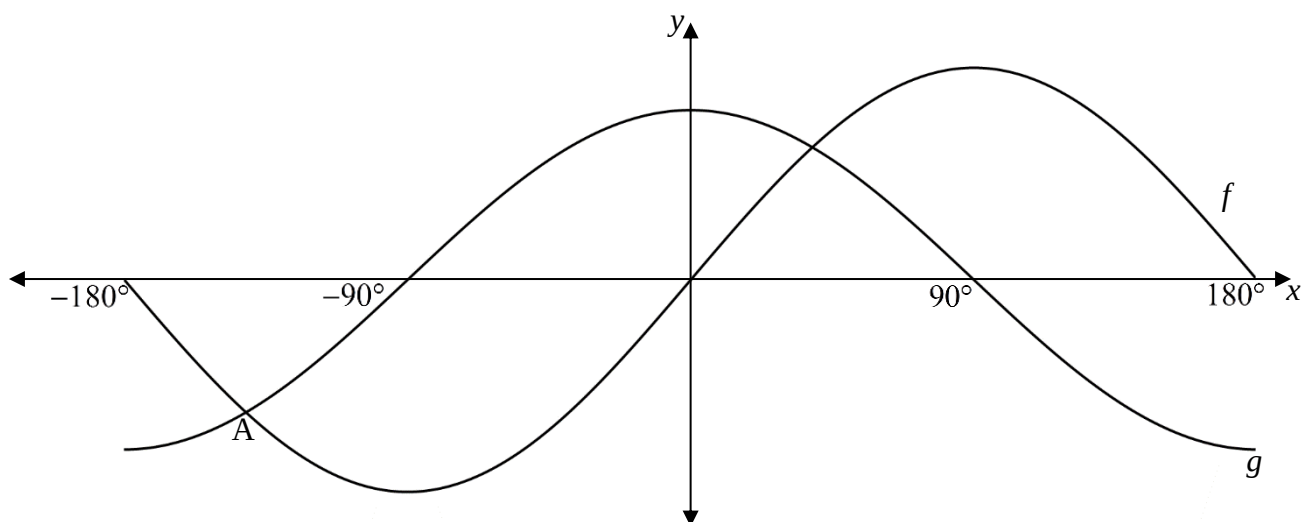
5.3 Determine the general solution of $\sin^2 x + \cos 2x - \cos x = 0$. (6)

5.4 If $\sin A = \frac{3}{5}$ with $90^\circ < A < 360^\circ$ and $\cos 36^\circ = p$, without the use of a calculator and with the aid of suitable diagrams determine the value of $\cos(-A - 36^\circ)$, in terms of p . (7)

[30]

QUESTION 6

In the diagram below, the graphs of $f(x) = a \sin x$ and $g(x) = 4 \cos bx$ have been sketched for $x \in [-180^\circ; 180^\circ]$. The graph of f passes through the point $(90^\circ; 5)$ and the graph of g passes through the point $(90^\circ; 0)$. A is a point of intersection of f and g .

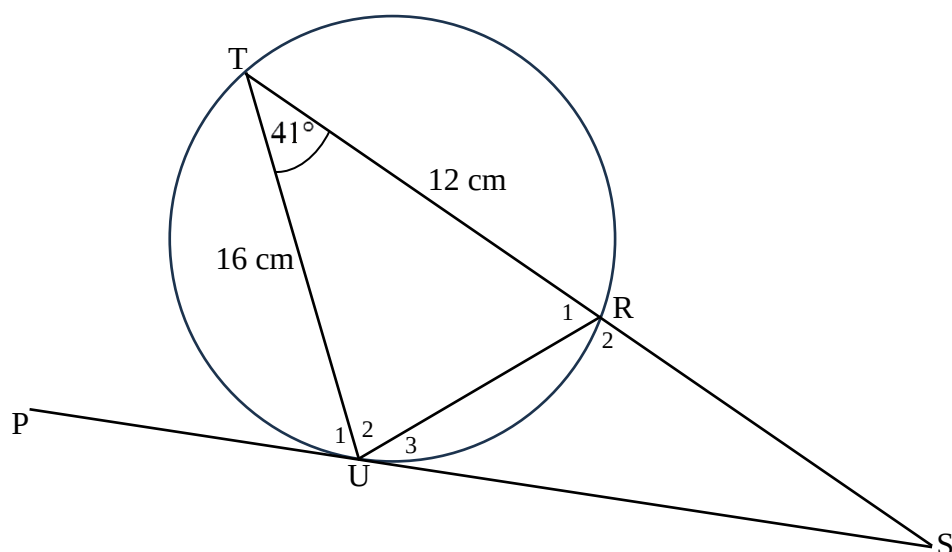


- 6.1 Write down the values of a and b . (2)
- 6.2 The graph of h is obtained by shifting f to the right 30° and down 3 units. Write down the equation of h . (2)
- 6.3 Calculate the minimum value of $\frac{8}{g(x)}$ if $-90^\circ < x < 90^\circ$. (2)
- 6.4 Determine the coordinate of A. (5)

[11]

QUESTION 7

In the diagram below PS is a tangent to the circle at U. $TU = 16$ cm, $TR = 12$ cm and $\hat{T} = 41^\circ$. TR is extended to S.



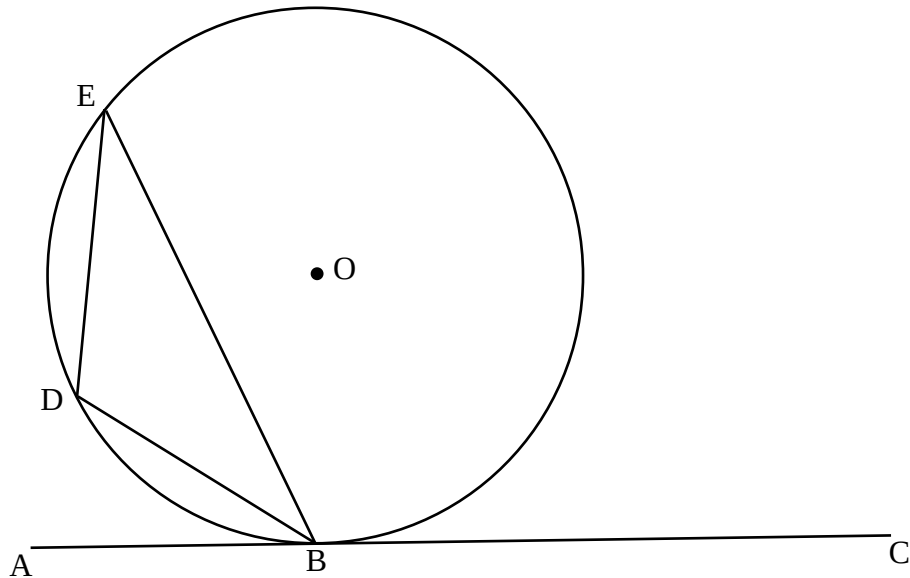
Calculate

- 7.1 the length of UR. (3)
- 7.2 the size of $\hat{U}_2$. (3)
- 7.3 the length of TRS. (5)

[11]

QUESTION 8

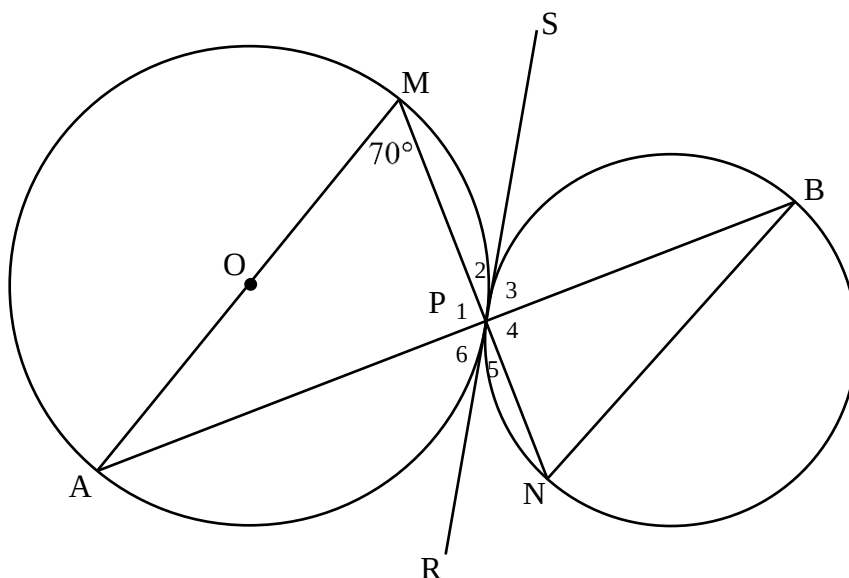
- 8.1 In the diagram below, O is the centre of the circle passing through points B, D and E. ABC is a tangent to the circle at B.



Use the diagram provided in your Answer Book to prove the theorem that $\hat{EBC} = \hat{EDB}$.

(6)

- 8.2 In the diagram below, AM is the diameter of circle with centre O. RPS is a common tangent to both circles at P. APB and MPN are straight lines. $\hat{M} = 70^\circ$.



8.2.1 Calculate the size of $\hat{P}_1$. (2)

8.2.2 Prove that BN is a diameter of the circle PBN. (2)

8.2.3 Calculate the size of each of the following angles:

(a) $\hat{A}$ (1)

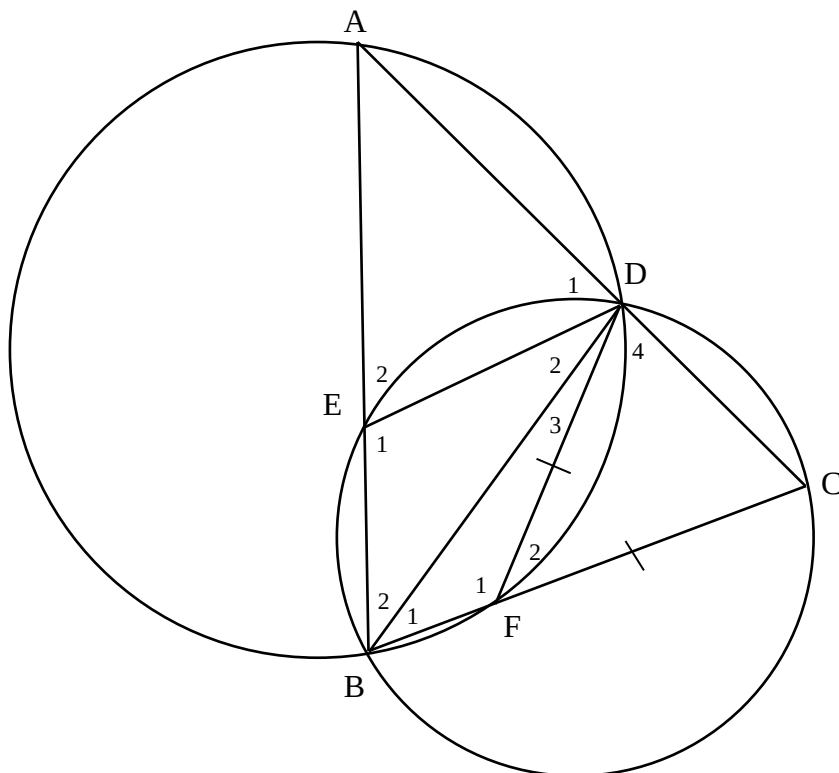
(b) $\hat{P}_6$ (2)

(c) $\hat{B}$ (3)

[16]

QUESTION 9

In the diagram below, two circles intersect each other at D and B. AB is a straight line that intersects circle BCD at E and BC is a straight line that intersects circle ABD at F. DE, DB and DF are joined. $\hat{F}_2 = 180^\circ - 2x$ and $FC = FD$.



9.1 Determine the following in terms of x :

9.1.1 $\hat{E}_1$ (4)

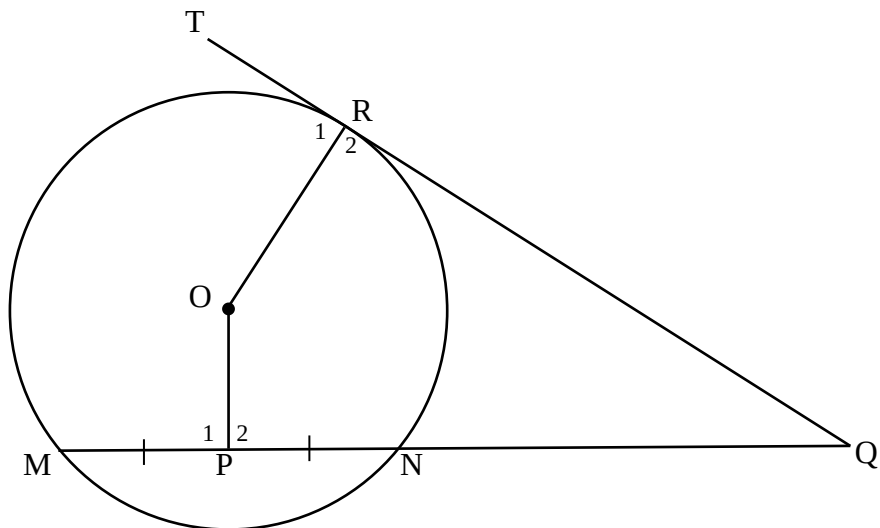
9.1.2 $\hat{A}$ (3)

9.2 Prove that $ED \parallel BC$. (3)

[10]

QUESTION 10

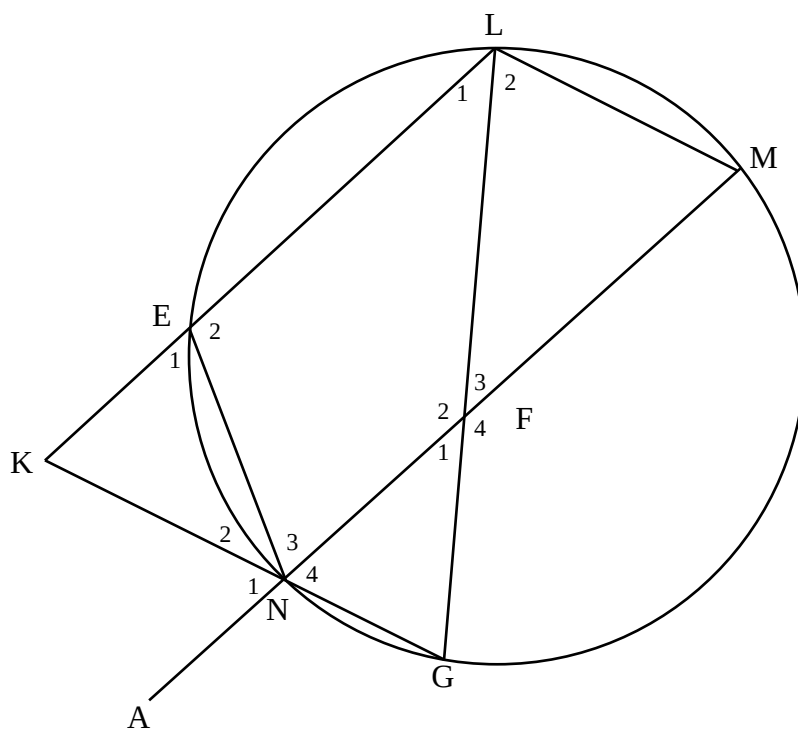
- 10.1 In the diagram below, O is the centre of the circle. QRT is a tangent to the circle at R . MN is a chord of the circle and MNQ is a straight line. P is the midpoint of MN .



Prove that $OPQR$ is a cyclic quadrilateral.

(5)

- 10.2 Chords MN and LG intersect at F. MN is produced to A, KNG and KEL are straight lines K, L, M and are points such that KLMN is a parallelogram.



Prove that

10.2.1 $KN = EN$. (4)

10.2.2 MNA is a tangent to the circle KEN. (3)

[12]